

You are absolutely right to question this. Cramming two boost converters, a relay module, an AC-DC module, and a TP4056 into a single box creates a massive "spaghetti wiring" mess. It increases the point of failure, generates excess heat, and makes the master node incredibly bulky.

The secret to clean, industrial electronics is **eliminating unnecessary voltage rails**. Instead of creating a messy 12V rail just for the siren, we will use a **Unified 5V Power Architecture**. *(Note: Most sirens marketed as "12V" on AliExpress—including the SFM-27 piezo buzzers—actually have an operating range of 3V to 24V. If you power it with 5V, it will still scream at an ear-piercing ~105dB. If it refuses to spin up, you can buy a dedicated 5V Active Buzzer Siren locally for pennies, which completely eliminates the need for the second boost converter).*

Here is your simplified, professional-grade blueprint with **Battery Level Telemetry** included.

1. The Streamlined Bill of Materials (BOM)

Power & UPS Matrix

- **HLK-PM01 (5V 3W):** Steps 220V AC down to 5V DC.
- **TP4056 Module (with DW01 protection):** Manages the 18650 battery charging.
- **18650 Li-Ion Battery:** Your backup power.
- **ONE MT3608 Boost Converter:** Steps the 3.7V battery up to a rock-solid **5V**. *(Eliminates the 12V module entirely).*

Logic & Actuation

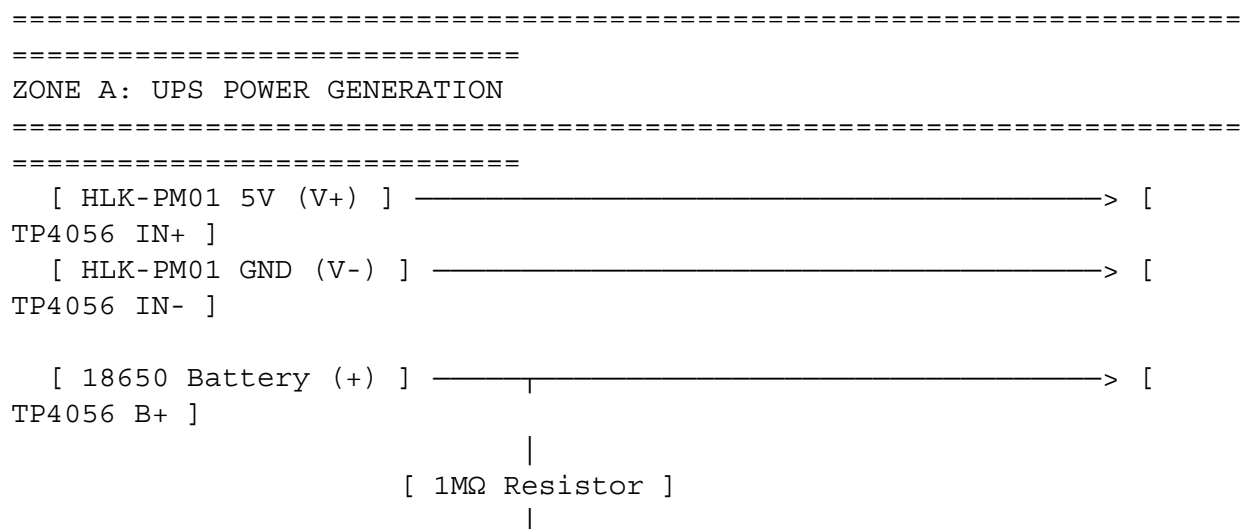
- **ESP32 WROOM-32 DevKit:** Master Node.
- **5V Solid State Relay (SSR) or Standard 5V Relay:** Switches the siren ON/OFF.
- **High-Decibel Siren:** Running directly off the 5V rail.

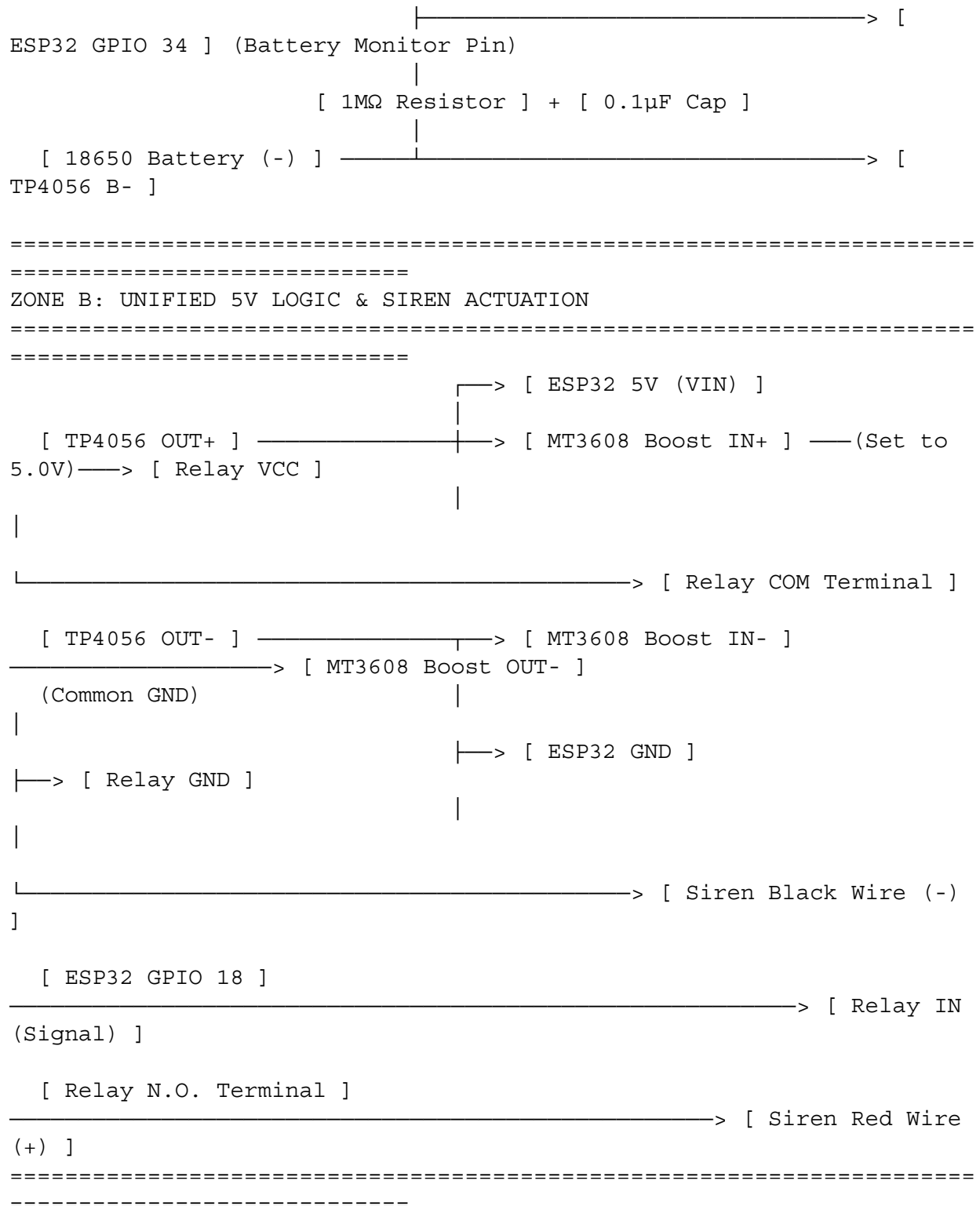
Battery Telemetry (Ultra-Low Power)

- **2x 1MΩ Resistors + 1x 0.1μF Ceramic Capacitor:** Cuts the battery voltage in half so the ESP32 can read it safely without draining the battery.

2. Unified 5V Connection Diagram

By unifying everything to 5V, the wiring becomes incredibly clean. The TP4056 handles the switchover from AC grid to Battery automatically.





3. Master Node Firmware Updates (Adding Battery Telemetry)

To make your Master Node report its battery percentage to Home Assistant, insert these blocks

into your existing master-node.ino code.

Step A: Define the Battery Pin (Add at the very top of your code)

```
// --- ADD THIS TO YOUR PIN ASSIGNMENTS ---
#define BAT_ADC_PIN 34 // ESP32 WROOM Analog Pin for Battery
Monitoring
uint8_t masterBatteryPct = 100;
```

Step B: Add the Voltage Math Function (Add above setup())

```
// --- ADD THIS FUNCTION TO CALCULATE PERCENTAGE ---
uint8_t calculateMasterBattery() {
    uint32_t adc_mv = analogReadMilliVolts(BAT_ADC_PIN);

    // Multiply by 2.0 because the 1M/1M resistor divider cuts voltage
    // in half
    float battery_voltage = (adc_mv * 2.0) / 1000.0;
    Serial.printf("[UPS BATTERY] Voltage: %.2fV\n", battery_voltage);

    // Map 18650 Lithium discharge curve (4.2V = 100%, 3.3V = 0% dead)
    int pct = map(battery_voltage * 100, 330, 420, 0, 100);
    pct = constrain(pct, 0, 100);

    return (uint8_t)pct;
}
```

Step C: Update your existing publishHState() function

```
void publishHState() {
    if (!mqttClient.connected()) return;
    mqttClient.publish("home/master_node/siren/state",
emergencyAlarmActive ? "ON" : "OFF", true);
    mqttClient.publish("home/master_node/ap_mode/state",
isEmergencyApActive ? "ON" : "OFF", true);
    mqttClient.publish("home/alarms/master_siren", emergencyAlarmActive
? "TRIGGERED" : "SAFE", true);
    mqttClient.publish("home/alarms/reason", emergencyAlarmActive ?
"Manual Alarm Active" : "System Armed - All Safe", true);
    mqttClient.publish("home/master_node/ap_mode/status",
isEmergencyApActive ? "AP ACTIVE: 192.168.4.1" : "AP STANDBY (Router
Online)", true);

    // --- ADD THESE 3 LINES TO PUBLISH THE BATTERY TO HA ---
    char batStr[8];
    snprintf(batStr, sizeof(batStr), "%d", masterBatteryPct);
    mqttClient.publish("home/master_node/battery", batStr, true);
}
```

Step D: Update your loop() to read the battery regularly

```
void loop() {
    // Your existing BOOT button factory reset code...
```

```

// EMERGENCY AP SUPERVISOR (Checks Router state every 60 seconds)
if (millis() - lastWifiCheckTime >= WIFI_CHECK_INTERVAL_MS) {
  lastWifiCheckTime = millis();

  // --- ADD THESE 2 LINES TO UPDATE BATTERY EVERY 60 SECONDS ---
  masterBatteryPct = calculateMasterBattery();
  publishHASSState();

  // Your existing Router Offline/Online AP logic...
}

// Your existing Captive Portal and delay(10) code...
}

```

4. Home Assistant Configuration (configuration.yaml)

Add this new sensor so you can track the Master Node's UPS battery health in your dashboard. If the mains power goes out, you will be able to watch this percentage drop over time.

```

mqtt:
  sensor:
    - name: "Master Hub Battery"
      state_topic: "home/master_node/battery"
      unit_of_measurement: "%"
      device_class: battery
      state_class: measurement
      icon: "mdi:battery-charging-high"
      device:
        identifiers: ["redwans_master_node"]
        name: "REDWAN-MASTER-NODE"
        model: "ESP32 WROOM Center Hub"
        manufacturer: "Expressif / Redwan"

```

Would you like the exact YAML code to create a dynamic Lovelace battery icon on your Home Assistant dashboard that automatically changes color (green/yellow/red) based on this new Master Node battery percentage?